

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks:30

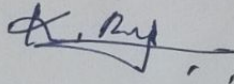
Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I K. Reshma (192585062) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



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= 90

NP

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CSA19174

Artificial

Intelligent

Job Interview Outcome - Resolution Algorithm

① Represent the statements in propositional logic

Let,

P = Farhan passed the interview.

H = Farhan is hired.

Given,

1. If an applicant passes the interview, then the applicant is hired.

$$P \rightarrow H$$

2. Farhan passes the interview

God: prove that Farhan is hired

H

② Convert statements into CNF

The implication is

$$P \rightarrow H$$

using:

$$P \rightarrow H \equiv \neg P \vee H$$

So the CNF clauses are

clause 1: $\neg P \vee H$

clause 2: P

Negate the goal

Goal:

Negated Goal:

H

$\neg H$

Therefore, the clauses used for resolution are:

1. $\neg P \vee H$

2. P

3. $\neg P$

③ Apply Resolution Algorithm step by step

step 1: Resolve clause 1 and clause 2.

clauses:

$\neg P \vee H$

P

Resolve on P :

$(\neg P \vee H), P$

Since P and $\neg P$ cancel:

H

New clause:

$\boxed{H}$

Step 2: Resolve the new clause with the negated goal

New clause:

Negated goal:

H

$\neg H$

Resolve:

$H, \neg H$

This produces

$\boxed{\quad}$

where $\boxed{\quad}$ represents the empty clause.

(1) Resolution proof

$$C_1 : \neg P \vee H$$

$$C_2 : P$$

$$C_3 : \neg H$$

Resolution:

$$C_1 + C_2 \Rightarrow H$$

Then:

$$H + C_3 \Rightarrow \boxed{\square}$$

Since the empty clause is obtained, the negation of the goal leads to a contradiction.

Farhan is hired

Therefore, Farhan is hired is proved using
The Resolution Algorithm

1) Represent the statements in propositional

logic

Let,

A = Meena attended all classes,

C = Meena gets attendance credit.

Knowledge Base:

1. If a student attends all classes,
then the student gets attendance credit.

$$A \rightarrow C$$

2. Meena attended the classes.

A

Goal: Prove that Meena gets attendance credit.

C

2) Convert into CNF and Negate the Goal

Convert into implication;

$$A \rightarrow C \equiv \neg A \vee C$$

CNF clauses:

* Clause 1: $\neg A \vee C$

* Clause 2: A

Negate the goal:

* Goal : c

* Negated Goal : $\neg c$

clause for resolution:

clause

Formula

$\neg A \vee C$

C_1

A

C_2

$\neg C$

C_3

③ Apply the Resolution Algorithm

step 1: Resolve C_1 and C_2

$$(C \vee A \vee C), A \Rightarrow C$$

New clause;
C

step 2: Resolve the new clause with C_3

$$C, \neg C \Rightarrow \square$$

④ Resolution proof

$$C_1: \neg A \vee C$$

$$C_2: A$$

$$C_3: \neg C$$

$$C_4: C (C_1, C_2)$$

$$C_5: \neg C (C_4, C_3)$$

conclusion

The Empty clause is obtained, so the negated goal leads to a contradiction.

Therefore, Meena gets attached credit is proved.

③ Car Rental Eligibility - Resolution Algorithm

1. propositional logic

Let,

L = Suresh has a valid license.

R = Suresh can rent a car.

Given,

1. If a driver has a valid license, then the driver can rent a car.

$L \rightarrow R$

2. Suresh has a valid license

L

Goal?

R.

② CNF and Negde the Goal

converts

$L \rightarrow R$

into CNF:

$\neg L \vee R$

so the clauses are:

$C_1: \neg L \vee R$

$C_2: L$

$C_3: \neg R$ - negated Goal

③ Resolution step by step

step 1: Resolve C_1 and C_2 :

$(\neg L \vee R), L$

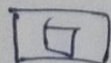
Therefore,

R

step 2: Resolve R with C_3 :

$R, \neg R$

Therefore,



① Conclusion.

The Empty clause is obtained.

Suresh can rent a car

Hence, the goal is proved using the Resolution Algorithm.

Supports

Achievement

Award.

4 Propositional Logic

Let:

S = Devya scored above 90%.

G = Devya won a gold medal

Given,

1. If a player scores above 90%, then the player won a gold medal.

$S \rightarrow G$

2. Devya scored above 90%.

S

Goal:

G

2) CNF and negate the Goal

convert:

$$S \rightarrow G$$

into CNF:

$$\neg S \vee G$$

clauses

$$C_1: \neg S \vee G$$

$$C_2: S$$

$$C_3: \neg G \rightarrow \text{negated Goal}$$

3) Resolution step by step.

step 1: resolve C_1 and C_2

$$(\neg S \vee G), S$$

Therefore

$$G$$

step 2: Resolve G with C_3 :

$$G, \neg G$$

Therefore,



conclusion

The Empty clause is obtained.

Dineya wins a gold medal

Hence, the goal is proved using the Resolution Algorithm.

Exam Hall Ticket Eligibility

1. propositional logic

A = student completed all assignment.

E = student is eligible for the exam.

H = student receives a hall ticket.

Knowledge bases

1. If a student completes all assignments, then the student is eligible for the exam.

$$A \rightarrow E$$

2. If a student is eligible for the exam, then the student receives a hall ticket.

$$E \rightarrow H$$

3. The student completed all assignments.
A

Goal:

H

2. convert into CMF

convert the implication.

$$A \rightarrow E \equiv \neg A \vee E$$

$$E \rightarrow H \equiv \neg E \vee H$$

So, the CMF clause are;

$$C_1: \neg A \vee E$$

$$C_2: \neg E \vee H$$

$$C_3: A$$

3. Negate the Goal

Goal:

H

Negated goal:

$\neg H$

Therefore, add;

$$C_4: \neg H$$

Resolution Algorithm

step 1: Resolve C_1 and C_3

$$(\neg A \vee E), A$$

cancel A and $\neg A$;

E

new clause:

$\boxed{E}$

step 2: Resolve E with C₂

$C_1: (\neg E \vee H)$

cancel E and $\neg E$;
H

New clause:

$\boxed{H}$

step 3: resolve H with the Negated Goal

H, $\neg H$

This gives:

$\boxed{}$

The empty clause is obtained

⑤ Final Resolution proof

$C_1: \neg A \vee E$

$C_2: \neg E \vee H$

$C_3: A$

$C_4: \neg H$

$$C_5 : \neg (C_1, C_3)$$

$$C_6 : H(C_5, C_2)$$

$$C_7 : \square (C_6, C_4)$$

Final conclusion

Since, the Empty clause is derived, The negation of the goal is contradictory.

The student receives a full tablet

Hence, The goal is proved using the Resolution Algorithm.